

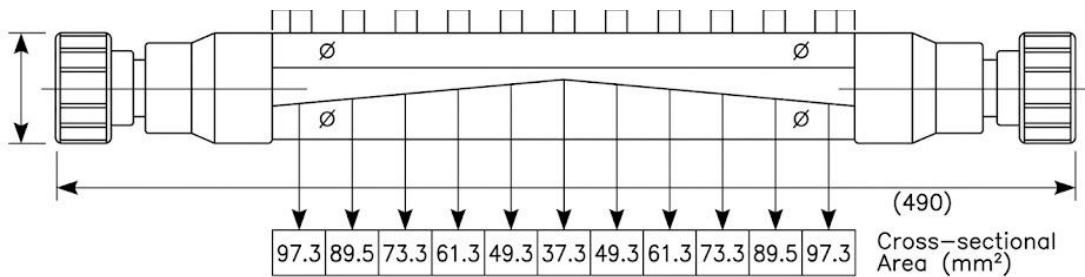
Exp.
02

Verification of Bernoulli's Equation

Bernoulli's Equation:

$$\frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2 + h_f$$

Cross Sectional Area:



Experimental data

No of observation	Volume of water collected v, liter	Time for collection t, sec	Flow rate $Q=v/t$, (m³/s)
1			
2			
3			

Tube position	Cross sectional area A, m²	Velocity $V=Q/A$ (m/s)	Velocity Head $V^2/2g$ (mm)	Piezometer Head $(\frac{P}{\gamma} + Z)$ mm	Total head mm
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					

Sample Calculation:

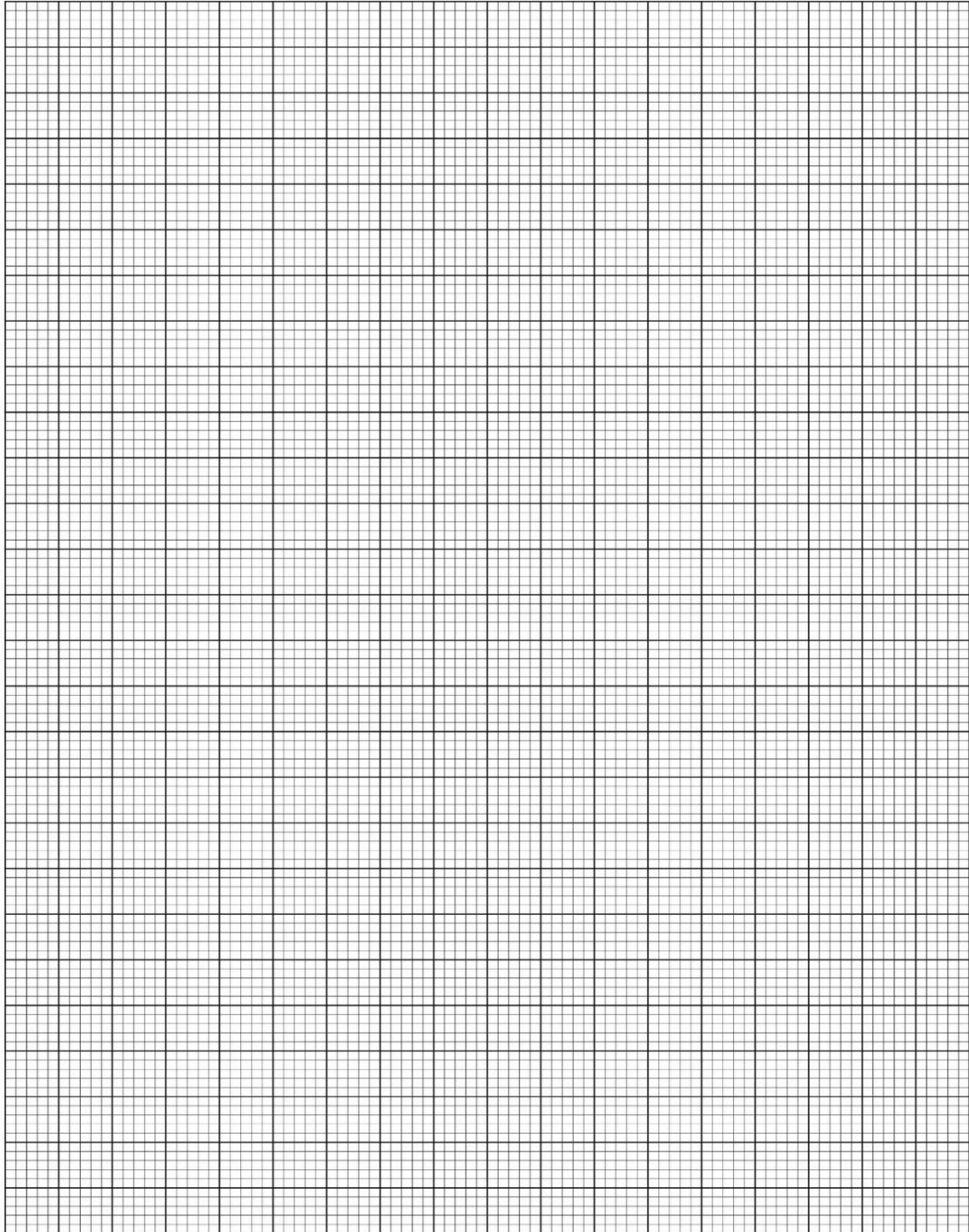
Show sample calculation for tube position **1** and **11**.

Plot

- Plot **Piezometer Head vs Tube Position, Velocity Head vs Tube Position and Total Head vs Tube Position**

Along Horizontal Axis:

Along Vertical Axis:



Discussion:

- Discuss whether Bernoulli's theorem is experimentally validated, with appropriate interpretation based on graphical results.
- Discuss why the total head does not remain constant along the flow, providing relevant physical reasoning.
- Identify the possible sources of measurement errors and discuss their effects on the experimental results.